Yiliu (Aiden) Li

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Authorized to work in the U.S. and China; no visa sponsorship required.

Education

Northwestern University Expected Jun 2027
M.S. in Computer Science (MSCS)

University College London Sep 2023 – Jul 2026
BSc Computer Science
Dissertation **Supervisor:** Anthony Steed

Publications

- **Aiden Yiliu Li**, Nels Numan, and Anthony Steed*. "**Visual Agentic Memory: Enabling Online Long Video Understanding via Online Indexing, Hierarchical Memory, and Agentic Retrieval**." arXiv | Code
- Wee Joe Tan, Zi Rui Lucas Lim, Shashank Durgad, Karim Obegi, and **Aiden Yiliu Li***. "**Avenir-UX: Automated UX Evaluation via Simulated Human Web Interaction with GUI Grounding**." arXiv | Code
- Yiheng Tian, Jingnan Zhang, Yiven Zhu, Fei Xu, Hao Kan, Leenuja Kannan, Niyaa Meganathan, and **Aiden Yiliu Li***. "**From Advisor to Voting Teammate: Institutional Authority and Information Structures of AI Agents in Bounded-Rational Human Groups**." *CHI 2026 Workshop.* PDF
- **Aiden Yiliu Li**, Xinyue Hao, Shilong Liu*, and Mengdi Wang*. "**Avenir-Web: Human-Experience-Imitating Multimodal Web Agents with Mixture of Grounding Experts**." arXiv | Code
- **Aiden Yiliu Li**, Bizhi Yu, Daoan Lei, Tianhe Ren, and Shilong Liu*. "**Chain-of-Ground: Improving GUI Grounding via Iterative Reasoning and Reference Feedback**." arXiv
- Violet Yinuo Han, Ziru Wei, **Aiden Yiliu Li**, Chris Wu, and Alexandra Ion*. "**Objects with Arms: How Robot Embodiment Shapes Perception in Physical Collaboration**." *Proceedings of the 35th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2026).*
- Jing Xue, Madhankumar Vasudevan, **Aiden Yiliu Li**, Zhouyang Shen, James Hardwick, Martin Dechant, and Marianna Obrist*. "**KiTaFab: Kinaesthetic and Tactile Mid-air Haptics Fabric Experiences**."

Research & Teaching Experiences

Research Internship Mar 2026 – Present
Stanford University Vision and Learning Lab (SVL)
- Conducting research in Computer Vision and Machine Learning, focusing on physical intelligence and structured representations for visual reasoning.
- **Supervisor:** Prof. Jiajun Wu.
- **Mentor:** Yue Gao.

Research Internship Sep 2025 – Present
Princeton University AI for Accelerating Invention (AI2) Lab
- Worked with Dr. Shilong Liu and Prof. Mengdi Wang on agentic AI for web-use agents and GUI-grounded interaction, developing an open-source system achieving state-of-the-art performance among open-source methods.

Research Internship Jun 2025 – Sep 2025
Carnegie Mellon University Human Computer Interaction Institute (CMU HCII)
- Integrated VLMs and VLAs on robotic arms for physical intelligence and HRI research, and studied how embodiment and interaction shape perceptions of robotic collaborators (submitted to CHI 2026).
- **Supervisor:** Prof. Alexandra Ion.

Research Assistant May 2025 – Dec 2025
Pervasive HCI Group, Tsinghua University
- Developing a GUI-based LLM platform for low-cost collection, organization, and synthesis of scientific facets, alongside a cross-platform mobile web app for AI hemifacial spasm analysis using facial analysis.

Research Assistant Aug 2024 – Apr 2025
UCL Computer Science Multi-Sensory Devices (MSD) Group
- Developed a VR environment for fabric-touch perception and supported kinaesthetic-illusion and pseudo-haptics research through experiments, literature reviews, presentations, and research summaries.
- **Supervisor:** Prof. Marianna Obrist.

Teaching Assistant Sep 2024 – Present
- *UCL COMP0016 System Engineering*: Supervised multidisciplinary, industry-sponsored CS teams (e.g., Microsoft, Intel, IBM, Cisco), coordinating weekly progress, stakeholder communication, and risk management.
- *UCL ENGF0034 Design and Professional Skills I (Computer Science)*: Mentored students in Python, AI game-player development, debugging, internet protocols, and networking; provided technical guidance across AI/ML (LLMs, NLP, RAG / Agentic AI), HCI, CG / AR / VR, computer vision, and cloud / mobile, and advised on system architecture and end-to-end implementation.

- **Module Leaders:** Prof. Dean Mohamedally and Prof. Martin Benning.

Industrial & Leadership Experiences

Head of Nexus Lab Sep 2025 – Sep 2026

UCL Artificial Intelligence Society

- Directing Europe's largest student-led AI research lab, overseeing multi-pillar programmes and leading faculty / PhD-mentored teams from ideation to publication.
- Organized regular academic events to connect students and researchers, designed research roadmaps, and built structured pipelines guiding students from learning to active research participation.

Campus Ambassador Oct 2025 – Sep 2026

TRAE, ByteDance

- Advocated for the AI-powered coding IDE on campus, organised workshops, and fostered a community of student developers.

Y Combinator AI Startup School 16–17 June, 2025

- Selected to join an exclusive program convening leading AI founders, researchers, and investors, including Sam Altman, Dr. Andrew Ng, Prof. Fei-Fei Li, Prof. Chelsea Finn, Garry Tan, and Satya Nadella, to explore frontier developments in AI innovation and entrepreneurship.

AI Product Manager Intern Feb 2025 – Feb 2026

Ponder AI (ResearchFlow)

- Actively supporting initiatives such as Keyboard Shortcuts, Mind Map View, and Accuracy Optimization.
- Proposed, designed, and launched a multi-functional AI chatbox that supports in-contextual quotation.
- Assisted in structuring product roadmaps, gathering user feedback, and optimizing AI functionalities to enhance usability and user engagement.

Projects

Serif

AI-native research workspace

- Building an AI research agent for LaTeX projects, supporting source verification, citation management, document revision, editable figures, and compilation.
- **Website: serif.is**

Ageniti

- An SDK for building apps that can be called by agents.
- Defines an action once and exposes it across CLI, HTTP, MCP, OpenAI / AI SDK tools, React, and typed clients through a shared runtime.
- **Website: ageniti.dev**
- **GitHub: github.com/Ageniti/ageniti**

SightLinks – UCL IXN Project Oct 2024 – Mar 2025

- Collaborated with UCL GDI Hub, Wheelchair Alliance, and Soundscape to enhance accessibility solutions for disabled individuals.
- Trained machine learning models based on MobileNetV3 and YOLO11, and used georeference techniques for precise zebra crossing detection and geolocation from satellite imagery, with precision above 85%.
- Achieved the **Highest Score in the Cohort**.
- **Project page: students.cs.ucl.ac.uk/2024/group15**

CUDA Accelerated SDOT Kernel Development at UCR Supercomputing Lab Jun 2023 – Aug 2023

- Achieved a breakthrough in SDOT computation by optimizing GPU-accelerated kernels, enhancing efficiency, and minimizing data movement through CUDA programming and related optimization techniques.
- **Source code: github.com/yiliu-li/Optimized-Cuda-SDOT-Kernel-on-NVIDIA-Turing-GPUs**
- **Supervisor:** Prof. Zizhong (Jeffery) Chen.